

Name: Qazi Abulaala
Sap ID: 72389
Instructor: Prof. Muhammad Imran

TASK 1: POLYMORPHISM

Calculator.java	   Share	Run	Output
<pre>1 class Calculator { 2 3 void calculate() { 4 int x = 8, y = 12; 5 System.out.println("Default Sum: " + (x + y)); 6 } 7 8 void calculate(int a, int b) { 9 System.out.println("Sum of 2 integers: " + (a + b)); 10 } 11 12 void calculate(double a, double b) { 13 System.out.println("Sum of 2 doubles: " + (a + b)); 14 } 15 16 void calculate(int a, int b, int c) { 17 System.out.println("Sum of 3 integers: " + (a + b + c)); 18 } 19 20 public static void main(String[] args) { 21 22 Calculator obj = new Calculator(); 23 24 obj.calculate(10, 20, 30); 25 obj.calculate(); 26 obj.calculate(5, 6, 5); 27 } 28 }</pre>			<pre>Sum of 3 integers: 60 Default Sum: 20 Sum of 2 doubles: 11.0 Sum of 2 integers: 40 === Code Execution Successful ===</pre>

TASK 2: POLYMORPHISM TEST

```
1 class Test {  
2  
3     void display(int num) {  
4         System.out.println("Integer method called");  
5         System.out.println("Value: " + num);  
6     }  
7  
8     void display() {  
9         System.out.println("No-parameter method called");  
10    }  
11  
12    public static void main(String[] args) {  
13  
14        Test obj = new Test();  
15  
16        obj.display(10);  
17        obj.display();  
18    }  
19 }
```

```
Integer method called  
Value: 10  
No-parameter method called  
  
=== Code Execution Successful ===
```

TASK 3: SUBTRACT THROUGH POLYMORPHISM

```
1- class Calculator {
2
3- void subtract(int a, int b) {
4     System.out.println("Result (2 ints): " + (a - b));
5 }
6
7- void subtract() {
8     int x = 20, y = 7;
9     System.out.println("Result (default): " + (x - y));
10 }
11
12- void subtract(double a, int b) {
13     System.out.println("Result (double-int): " + (a - b));
14 }
15
16- void subtract(int a, int b, int c) {
17     System.out.println("Result (3 ints): " + (a - b - c));
18 }
19
20- public static void main(String[] args) {
21
22     Calculator obj = new Calculator();
23
24     int a = 12, b = 6, c = 3;
25
26     obj.subtract();
27     obj.subtract(a, b);
28     obj.subtract(7.5, 2);
29     obj.subtract(a, b, c);
30 }
```

Result (default): 13
Result (2 ints): 6
Result (double-int): 5.5
Result (3 ints): 3

=== Code Execution Successful ===

TASK 4: POLYMORPHISM SUBTRACT AND ADD

```
class Calculator {  
  
    void sum() {  
        int x = 15, y = 25;  
        System.out.println("Sum (default): " + (x + y));  
    }  
  
    void sub() {  
        int x = 30, y = 10;  
        System.out.println("Sub (default): " + (x - y));  
    }  
  
    void sum(int a, int b) {  
        System.out.println("Sum (2 numbers): " + (a + b));  
    }  
  
    void sub(int a, int b) {  
        System.out.println("Sub (2 numbers): " + (a - b));  
    }  
  
    public static void main(String[] args) {  
  
        int a = 50;  
        int b = 20;  
  
        Calculator obj = new Calculator();  
  
        obj.sum();  
        obj.sub();  
        obj.sum(a, b);  
        obj.sub(a, b);  
    }  
}
```

```
Sum (default): 40  
Sub (default): 20  
Sum (2 numbers): 70  
Sub (2 numbers): 30
```

--- Code Execution Successful ---